## 1.1 EM Initialization

Which solution does the EM algorithm converge to if we initialize the algorithm with identical parameters for all clusters: $\mu_1^{(0)} = \cdots = \mu_K^{(0)}$ and $\Sigma_1^{(0)} = \cdots = \Sigma_K^{(0)}$?

Lets denote the common initial parameters by $\mu^{(0)}$ and $\Sigma^{(0)}$.

**E-step**

The E-step computes $r_{ik}^{(t)}$, defined as:

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i|\mu_j, \Sigma_j)} \tag{1}$$

Given that $\mu_k = \mu^{(0)}$ and $\Sigma_k = \Sigma^{(0)}$ for all $k$, the likelihood term $\mathcal{N}(x_i|\mu_k, \Sigma_k)$ is identical for all clusters.
Just for ease of reading, lets denote this common term as $L_i = \mathcal{N}(x_i|\mu^{(0)}, \Sigma^{(0)})$.

Substituting $L_i$ into $r_{ik}$ equation:

$$r_{ik} = \frac{\pi_k L_i}{\sum_{j=1}^{K} \pi_j L_i} = \frac{\pi_k L_i}{L_i \sum_{j=1}^{K} \pi_j} \tag{2}$$

Since $\sum_{j=1}^{K} \pi_j = 1$, $r_{ik}$ simplifies to:

$$r_{ik} = \pi_k \tag{3}$$

This implies that the posterior probability of a data point belonging to cluster $k$ is independent of the data point $x_i$ itself. It relies only on the prior $\pi_k$.

**M-step**

For updating the means $\mu_k^{(t+1)}$:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{N} r_{ik} x_i}{\sum_{i=1}^{N} r_{ik}} \tag{4}$$

Substituting $r_{ik} = \pi_k$:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{N} \pi_k x_i}{\sum_{i=1}^{N} \pi_k} = \frac{\pi_k \sum_{i=1}^{N} x_i}{N\pi_k} = \frac{1}{N} \sum_{i=1}^{N} x_i = \bar{x} \tag{5}$$

Similarly, for the covariance matrices $\Sigma_k^{(t+1)}$:

$$\Sigma_k^{(t+1)} = \frac{\sum_{i=1}^{N} \pi_k (x_i - \mu_k)(x_i - \mu_k)^T}{N\pi_k} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})(x_i - \bar{x})^T = S \tag{6}$$

where $\bar{x}$ is the sample mean and $S$ is the sample covariance of the entire dataset.

## Results

The EM algorithm converges to a solution where all $K$ components are identical.

- All means $\mu_k$ converge to the global mean of the dataset.
- All covariance matrices $\Sigma_k$ converge to the global covariance of the dataset.

Because the components remain identical at every step, the algorithm isn't able to separate the data into distinct clusters.

What we get is that the algorithm fits the entire data set into a single Gaussian distribution.

## 1.2 Gibbs Sampling

Find the conditional distributions $p(x|y)$ and $p(y|x)$ needed for the Gibbs sampling algorithm under the given joint distribution:

$$p(x,y) = \frac{1}{Z}\mathcal{N}(x|\mu_x,\Sigma_x)\mathcal{N}(y|\mu_y,\Sigma_y)e^{-\frac{\beta}{2}\|x-y\|^2} \tag{7}$$

The Gibbs sampling algorithm, iteratively samples from the conditional distributions of each variable while holding the others fixed.

**Calculating** $p(x|y)$

From the definition of conditional probability, $p(x|y) \propto p(x,y)$ when $y$ is fixed (treated as a constant).
We remain only with the terms in the joint distribution that depend on $x$:

$$p(x|y) \propto \mathcal{N}(x|\mu_x,\Sigma_x)\cdot e^{-\frac{\beta}{2}\|x-y\|^2} \tag{8}$$

From properties we have seen previously, the resulting conditional distribution will also be Gaussian.
Let $p(x|y) = \mathcal{N}(x|\hat{\mu}_x,\hat{\Sigma}_x)$.
To find the parameters $\hat{\mu}_x$ and $\hat{\Sigma}_x$, we will look at the log-density $\ln p(x|y)$ and we will complete the square.

When ignoring anything independent of $x$, if we expand the log-density:

$$\begin{aligned}\ln p(x|y) &\propto -\frac{1}{2}(x-\mu_x)^T\Sigma_x^{-1}(x-\mu_x) - \frac{\beta}{2}(x-y)^T(x-y)\\ &= -\frac{1}{2}\left[x^T\Sigma_x^{-1}x - 2x^T\Sigma_x^{-1}\mu_x\right] - \frac{\beta}{2}\left[x^Tx - 2x^Ty\right] + \text{const}\end{aligned}$$

And group the terms that are quadratic and linear in $x$:

$$\ln p(x|y) \propto -\frac{1}{2}x^T\left(\Sigma_x^{-1}+\beta I\right)x + x^T\left(\Sigma_x^{-1}\mu_x + \beta y\right) + \text{const} \tag{9}$$

When comparing what we got with the standard form of a Gaussian log-density:

$$\ln\mathcal{N}(x|\hat{\mu},\hat{\Sigma}) \propto -\frac{1}{2}x^T\hat{\Sigma}^{-1}x + x^T\hat{\Sigma}^{-1}\hat{\mu} \tag{10}$$

By comparing the quadratic and linear terms we will find the mean and covariance.

**The Covariance:**

$$\hat{\Sigma}_x = \left(\Sigma_x^{-1} + \beta I\right)^{-1} \tag{11}$$

**The Mean:**

$$\hat{\mu}_x = \hat{\Sigma}_x \left(\Sigma_x^{-1}\mu_x + \beta y\right) \tag{12}$$

**Calculating** $p(y|x)$

Following the same derivation steps for $y$ (treating $x$ as fixed), we obtain a Gaussian distribution $\mathcal{N}(y|\hat{\mu}_y, \hat{\Sigma}_y)$ with parameters:

$$\hat{\Sigma}_y = \left(\Sigma_y^{-1} + \beta I\right)^{-1} \tag{13}$$

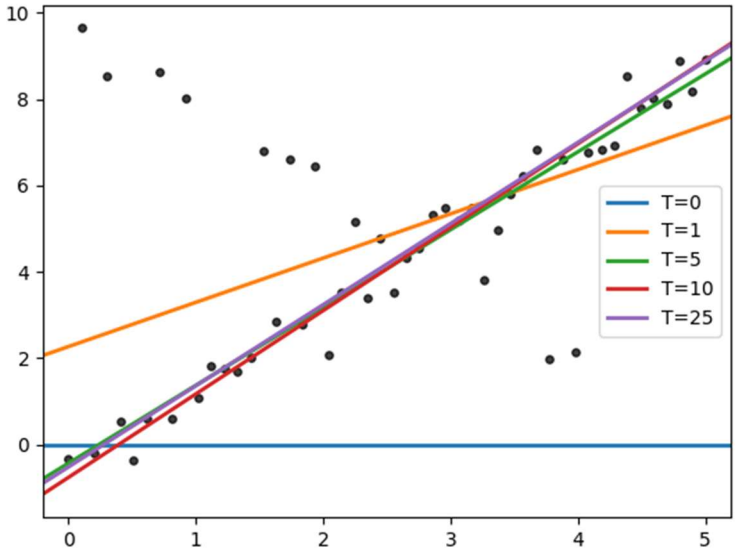
$$\hat{\mu}_y = \hat{\Sigma}_y \left(\Sigma_y^{-1}\mu_y + \beta x\right) \tag{14}$$

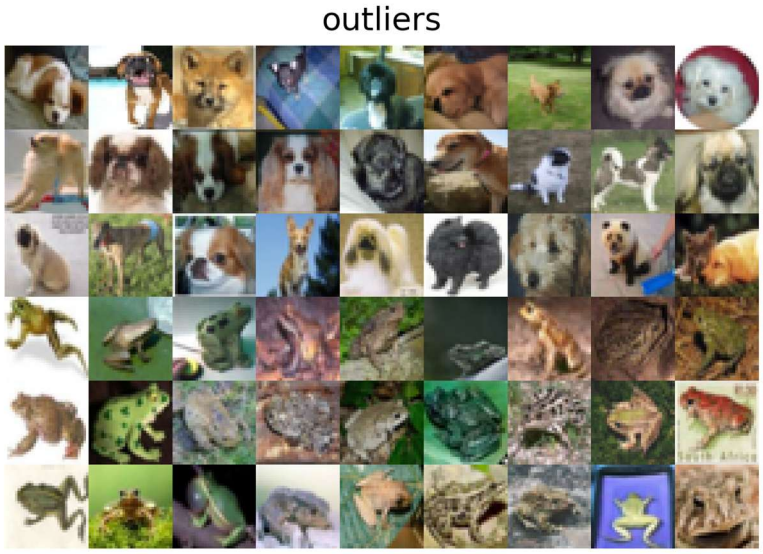
## Results

The conditional distributions required for Gibbs sampling are:

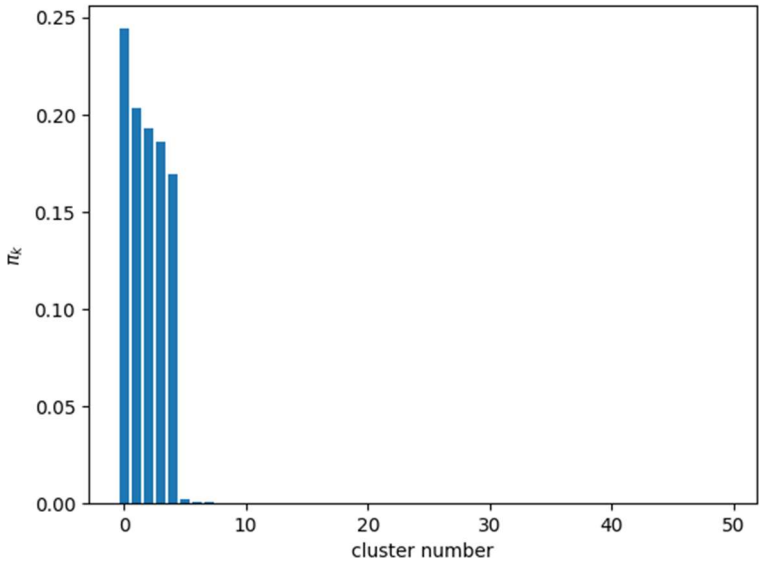
- $p(x|y) = \mathcal{N}(x|\hat{\mu}_x, \hat{\Sigma}_x)$
- $p(y|x) = \mathcal{N}(y|\hat{\mu}_y, \hat{\Sigma}_y)$

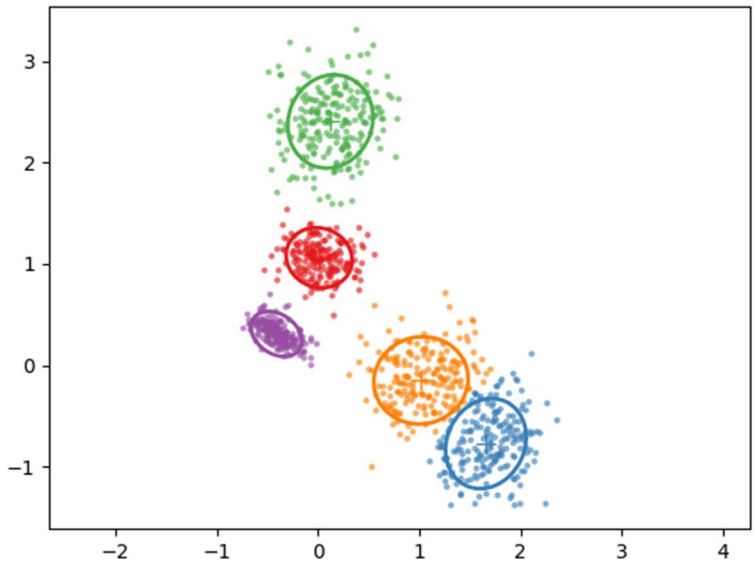
2 – Robust Regression

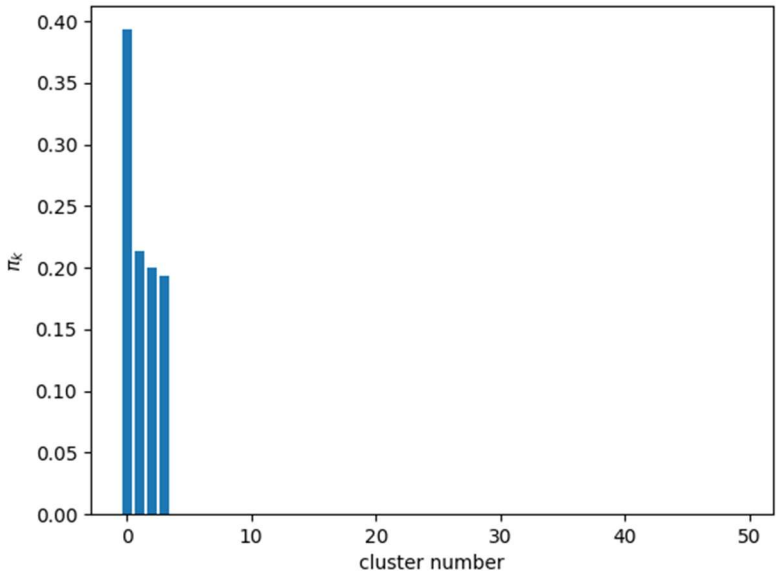


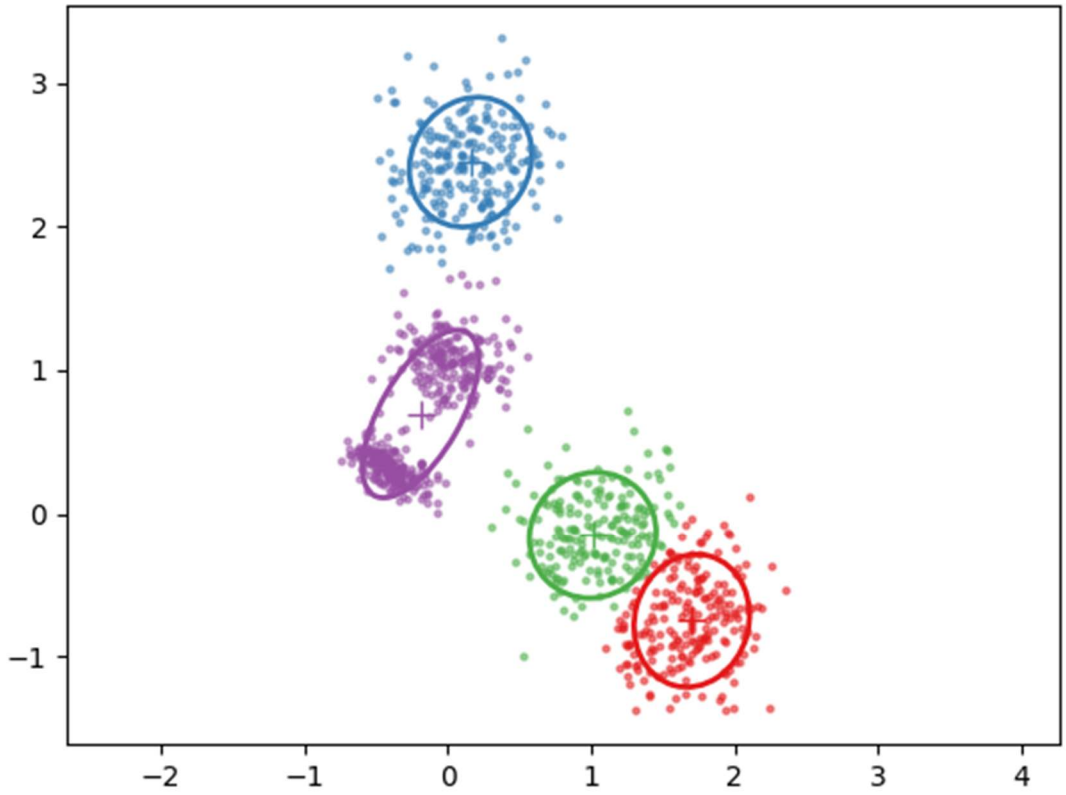



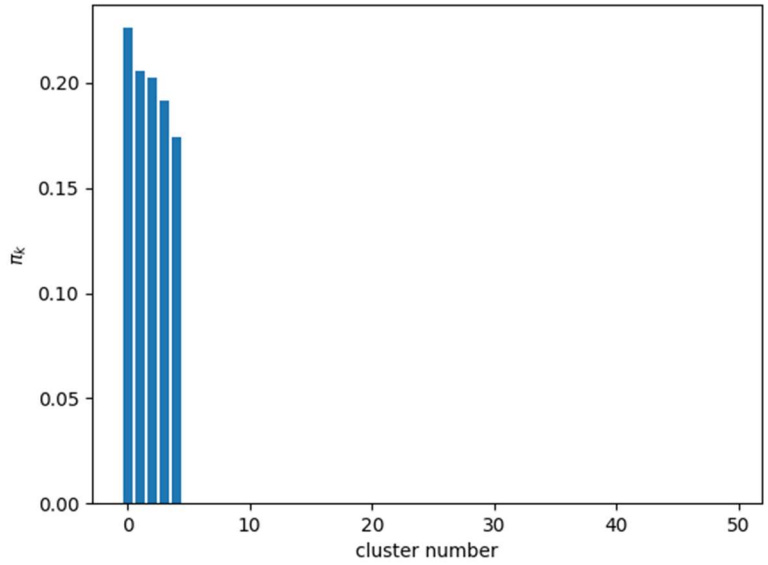
3 – Bayesian GMMs

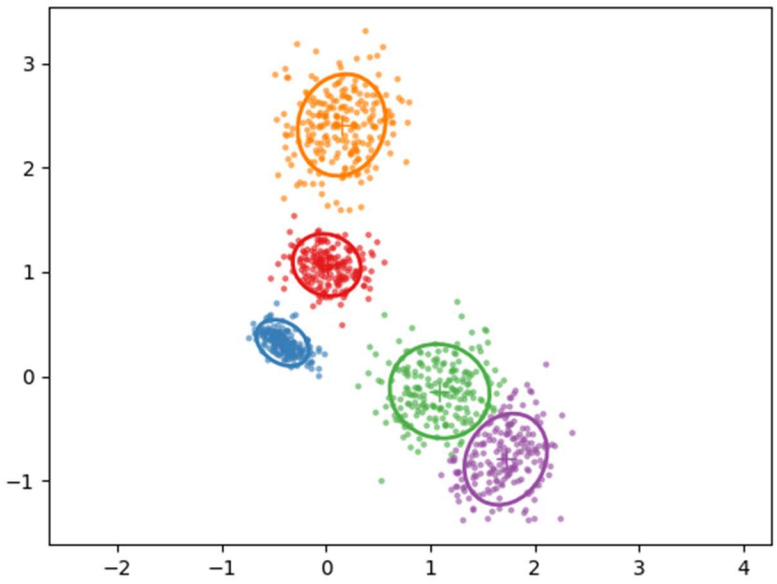


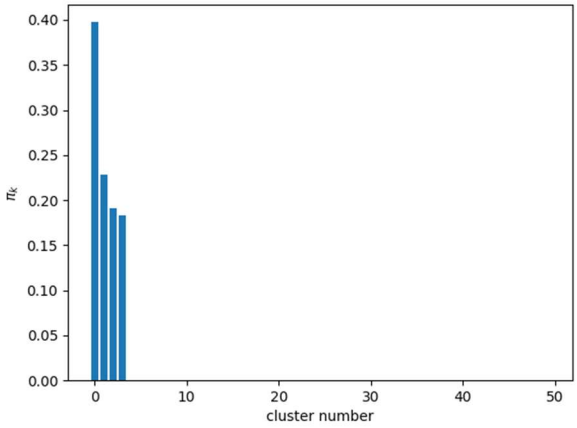



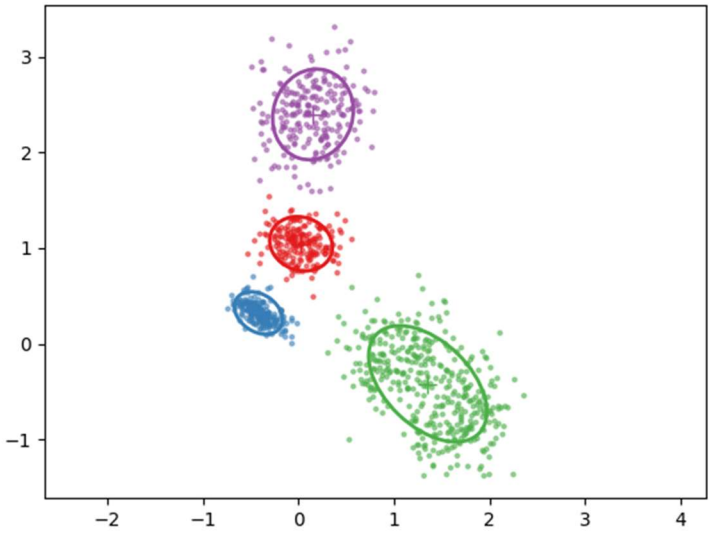

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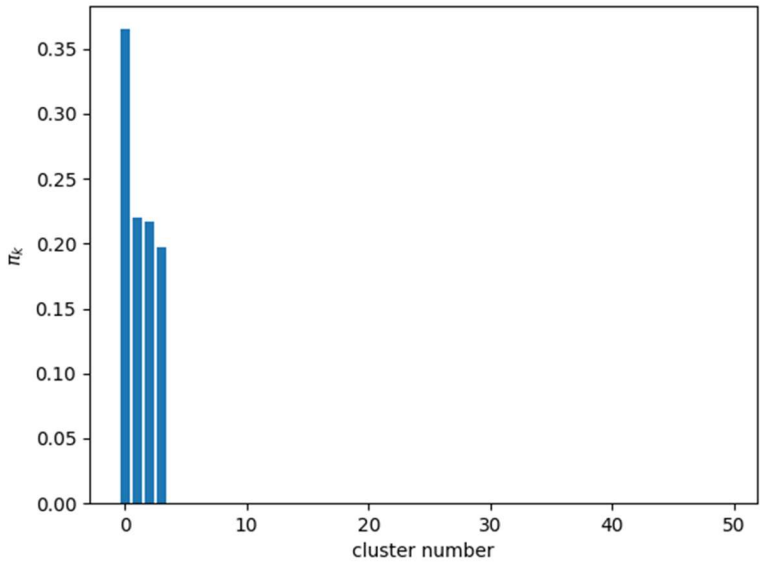

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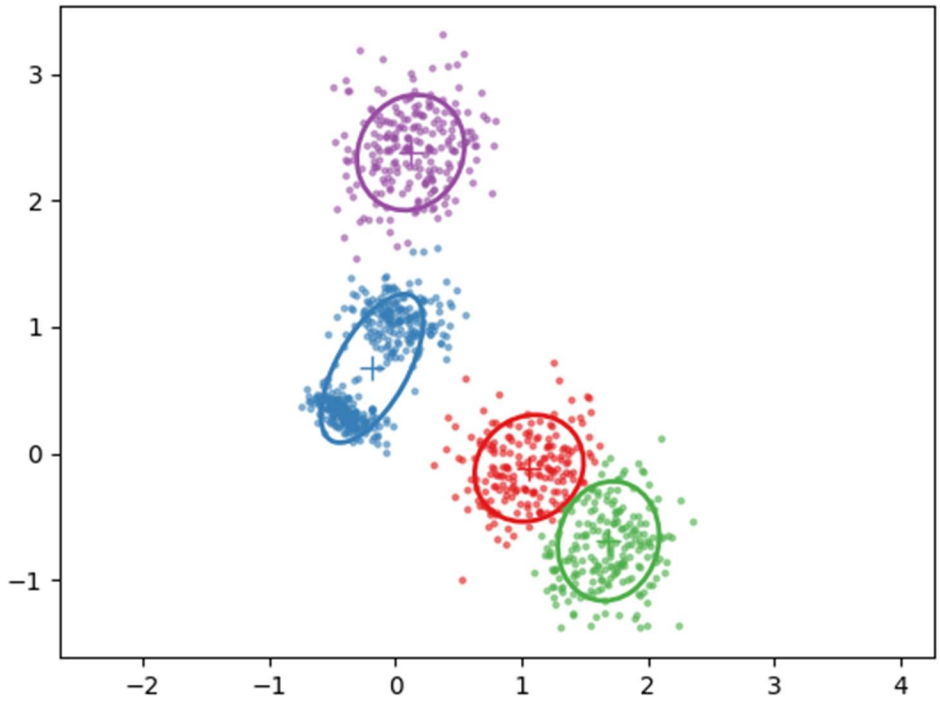

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